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A Broad Examination on Bacterial Responses to a Wide Range of Biocide Chemicals and Exploration of Potential 'Biosensor' Strategy

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Declaration

1. Raw chemical material of this research was provided by Kiran Patil's laboratory (Cambridge, UK).
2. Majority of the data was collected by the author, Zhe Zhang. A small proportion of raw data (growth curve data of 3 out of 21 strains of bacteria) was provided by Thomas P. Smith.
3. Data cleaning, processing, and analysing were performed by the author, Zhe Zhang, with help and instructions from Thomas Bell and Thomas P. Smith.
4. Any mathematical model or method used in this research has been cited correctly in the main text.
5. Both the supervisor, Thomas Bell, and co-supervisor, Thomas P. Smith provided key instructions and advice on experimentations and data analyses that have contributed greatly to this research.

Abstract

A large variety of pesticides has been increasingly used for agricultural purposes globally and is causing severe ecological crises. Non-target effects of pesticides on organisms, especially microbes which are vital for ecosystem functioning, must be deeply studied in order to better investigate the impact of pesticide uses on ecosystems. Current knowledge of bacterial responses to pesticide chemical pollutants is nearly entirely based on our understanding of typical laboratory-use 'model' strains of bacteria and case studies or experiments focused on specific chemicals and bacterial species. In real world ecosystems, bacterial species under stresses of pesticide chemical pollutants have a much higher phylogenetic diversity, and the pesticide chemicals have a much wider spectrum. Here I generally assayed the responses of 21 strains of bacteria across various phylogenetic taxonomies to 168 different chemicals which are major ingredients of typical pesticides. Through this research, I have concluded two major findings on how pesticide chemicals and bacteria may interact. Chemicals classified by their pesticidal target organisms do not show similarity in the way they affect bacteria, while phylogenetically related bacterial species are evidenced to react similarly to the pesticide chemicals. The outcoming of this research gives implications for impact of leaching pesticides, and provides a possible direction for future 'biosensor' exploration. In possible further studies of potential bacterial biosensors, chemicals should be classified based on their effect on bacteria instead of original groups of disinfectant targets; in the perspective of bacteria, the response to chemicals may be inferred by that of a phylogenetically similar species.

1. Introduction

Agricultural disinfectants including pesticides have been widely and increasingly used all over the continents for decades: global annual agricultural pesticide usage has risen from 1.80 million tonnes of active ingredients by 1990 to 3.54 million tonnes by 2021, with an increase of 49% (Food and Agriculture Organization of the United Nations, 2023). Although it is necessary to use chemical pesticides in agriculture, the residues of many pesticides have been found affecting non-target organisms, mainly on soil microorganisms, plants or animals, and eventually human (Atlas *et al.*, 1978; Shao and Zhang, 2017; Zaller and Brühl, 2019). These non-target effects of pesticides have been evidenced extremely harmful to both ecosystems and human health (e.g., Margni *et al.*, 2002; Zhen *et al.*, 2019; INSERM Collective Expertise Centre, 2022). Much attention has been paid to study the impact of the chemical pollutants on ecosystems as their threats have been rising dramatically (Gautam and Anbumani, 2020) but many aspects have remained unclear, where their impacts on microbes can be a very important clue but also a poorly understood area.

One of the key research focuses is the effect on non-target microbes by pesticide chemicals leached into waters. During the water cycle, hydrological processes, and irrigations, pesticides had been found leaching into water resources including ground waters (Jablonowski *et al.*, 2012; Rosenbom *et al.*, 2015; Hochman *et al.*, 2021). Chemical residues of pesticides may escape from soil by precipitation and irrigation, then contaminate natural water resources during water infiltration and runoff from soil (Vogel *et al*., 2015; Zajíček *et al.*, 2018; Hochman *et al.*, 2021). Microbes as the most abundant living organisms and one of the major participants in ecological processes, plays a vital role in ecosystem functioning but under great anthropogenic threats like pesticide pollutions (Sorokin, 1973; Bell *et al*., 2005; Zhou *et al*., 2020). Current studies of non-target effects of pesticides on microbes have been mainly focused on soil-microorganisms (Johnsen *et al*., 2001; Imfeld and Vuilleumier, 2012; Dhanker *et al.*, 2021), however, large amount of pesticide chemicals have been leaching into waters annually and have been found reducing the biodiversity in the polluted waters (Singh *et al.*, 2021). In addition, because of the deep connection between water and human activities and the long-term hazard of pesticide chemicals to human health as mentioned, pesticide residues in water resources can potentially be a very severe global/regional health problem. Therefore, it is crucial to understand the impact on microbes resulted by leached chemical pollutants in the water, in order to further investigate the influence on ecosystems brought by pesticides.

Studies of bacterial responses to pesticide chemicals had been conducted over the past decades trying to indicate the ecological impact brought by pesticide pollution, while most of which are case studies based on specific strains/communities of bacteria or experiments with relatively small variety of pesticides (Muturi *et al*., 2017; Onwona-Kwakye *et al*., 2020; Chen *et al.*, 2021). The limited number and diversity of strains and chemicals resulted in unrepresentative indications, as ecosystems in real world have much higher complexity and broader diversity of organisms and are under stresses of a much wider range of chemicals. The bacteria that are used for testing the non-target effects of toxic chemicals have also been constrained to a few 'model' species, including *Aliivibrio fischeri*, *Vibrio anguillarum*, and *Escherichia coli* (Parvez *et al.*, 2006; Robbens *et al.*, 2010; Rotini *et al.*, 2017), which their responses to chemicals do very limited help to reveal the ecological

impact in real world scenarios due to the narrow diversity of these 'lab-use' model strains of bacteria (Smith *et al.*, 2023). This research is a laboratory based analytical experiment which covers different strains of bacteria across a wide range of phylogenetic diversity collected from unpolluted streams in Iceland, with a broad spectrum of pesticide chemical compounds. Ideally, I would like to find certain type of pesticide chemicals that have same effect on bacteria, and specific type of bacteria that react in the same way to pesticide chemicals. Therefore, focus of this research has been to determine whether chemicals of same disinfecting target group affect bacteria similarly, and conversely, whether evolutionarily close bacterial species react to chemicals similarly. By doing so, this research aims at: 1. Conclude the regular patterns of how different pesticide chemicals affect bacteria and how different bacteria are affected by chemicals, then gives natural inclination in classifications of chemicals and bacteria. 2. Preliminarily exploring, or at least providing clues to, possible bacterial indicators of pesticidal pollution which can be helpful in scenarios such as risk assessment and environmental monitoring.

Here I inspected population growth curves of 21 strains of bacterial species as their overall fitness, under the stress of 168 different chemical compounds which are major ingredients of all kinds of pesticides. Clustering analysis and PCA ordination were applied to identify partitions of chemicals by bacterial responses and compare with classification by target organisms. Further phylogenetic analysis was carried out using 16S Ribosomal DNA sequences of the bacteria. By comparing evolutionary relatedness of the strains with their corresponding reactions to chemical stresses (as the phenotypic responses) using mantel test, the correlation between phylogeny and chemical responses of bacteria is examined. There is few strain found by the end of this experiment that can be used as 'Biosensor' of certain type of pesticide straight away, but the results of this research can be useful clues and potential methods for future studies.

Conclusively, in this research I have proposed and examined the two hypotheses, that chemicals of pesticides with same target groups have similar impacts on bacteria, and phylogenetically similar bacteria have similar responses to pesticide chemicals. First hypothesis was examined by clustering analysis and PCA, while second was examined by phylogenetic analysis and Mantel test. The verification of the hypotheses indicatively gives useful threads for future applications on biosensing using bacteria.

2. Methodology

2.1. Experiment Overview. The research was carried out in two parts, bacteria growing & measuring and data analysis. In bacteria growing and measuring part, bacteria was grown, and the growth curves of bacteria prepared as the raw data for analyses were measured. In data analysis part, three further sections were included. Section one is to analyse the data of growth curves of bacterial strains. Section two is to verify the first hypothesis, consists of clustering analysis and principal component analysis. Section three is to verify hypothesis, consists of phylogenetic signal testing and Mantel test.

2.2. Bacteria Solution Preparation. 168 different chemicals provided by Kiran Patil's laboratory (MRC Toxicology Unit, Gleeson Building, Tennis Court Road, Cambridge, UK) were added into two 96-well assay plates, pesticide library 1 and 2. Each library has 3 replicates with different arrangements of chemicals to lessen the edge effect, which means chemical solutions at the edge of the 96-well plates evaporate at a higher rate than those at more central positions. Therefore, there were in total 6 combinations of 2 libraries and 3 replicates per strain being tested. The chemicals were chosen from commonly used pesticides that target certain types of organisms. The summary of target groups of the chemicals is stated in table 1.

Fungicide	Herbicide	Herbicide safener	Insecticide	Insecticide, Acaricide	Multi-target
49	41	1	53	10	7
Parasiticide	Plant Growth Regulator	Pro-insecticide	Rodenticide	Synergist	
1	2	1	1	2	

Table.1. Number of pesticide chemicals in each of the group by their disinfecting targets

Sequentially, 21 strains of bacteria were chosen from the library which the selection of bacterial species maximized the phylogenetic diversity, and the bacteria selected must grow normally under room temperature. Species names of all bacteria with their phylogenetic taxonomies are shown in the table 2 below. Bacteria were isolates originally collected from sediments of freshwater streams in Iceland (details see Smith *et al.*, 2023) and were provided from Thomas Bell laboratory's isolate library (Unit C, Silwood Park Campus, Imperial College London, Buckhurst Road, Ascot, SL5 7PY, UK). It is to make sure that the bacteria were isolated from relatively pristine environments and are genetically less affected or selected by stresses of anthropogenic pollutants at pre-experiment stage so that they react to chemicals primitively. The bacteria strains were inoculated in Luria Broth solution (L.B. solution, which is a nutrient rich growth medium) and left for grown for 4 days to reach carrying capacity. Afterwards bacterial solutions at carrying capacity were 1 in 50 times diluted by L.B. media and pipetted into each well of the chemical plates, for both pesticide libraries and all three replicates. This diluted the bacteria to 1 in 100 from carrying capacity, and the chemicals to the required 20μM concentration which is commonly used in high-throughput screening experimentations of drugs on bacteria (Maier *et al.*, 2018).

Strain.id	Phylum	Class	Order	Family	Genus	Species
565	Actinobacteria	Actinomycetia	Micrococcales	Intrasporangiaceae	Oryzobacter	Oryzobacter terrae
390	Actinobacteria	Actinomycetia	Micrococcales	Microbacteriaceae	Microbacterium	Microbacterium arborescens
527	Actinobacteria	Actinomycetia	Micrococcales	Micrococcaceae	Arthrobacter	Arthrobacter humicola
442	Bacteroidetes	Cytophagia	Cytophagales	Hymenobacteraceae	Hymenobacter	Hymenobacter glaciei
74	Firmicutes	Bacilli	Bacillales	Bacillaceae	Neobacillus	Neobacillus soli
186	Firmicutes	Bacilli	Bacillales	Paenibacillaceae	Paenibacillus	Paenibacillus castaneae
448	Firmicutes	Bacilli	Lactobacillales	Carnobacteriaceae	Carnobacterium	Carnobacterium gallinarum
436	Proteobacteria	Alphaproteobacteria	Caulobacterales	Caulobacteraceae	Brevundimonas	Brevundimonas staleyi
85	Proteobacteria	Alphaproteobacteria	Hyphomicrobiales	Boseaceae	Bosea	Bosea massiliensis 63287
371	Proteobacteria	Alphaproteobacteria	Hyphomicrobiales	Rhizobiaceae	Rhizobium	Rhizobium herbae
398	Proteobacteria	Alphaproteobacteria	Sphingomonadales	Sphingomonadaceae	Sphingobium	Sphingobium czechense LL01
88	Proteobacteria	Betaproteobacteria	Neisseriales	Neisseriaceae	Prolinoborus	Prolinoborus fasciculus
487	Proteobacteria	Gammaproteobacteria	Aeromonadales	Aeromonadaceae	Aeromonas	Aeromonas popoffii
374	Proteobacteria	Gammaproteobacteria	Aeromonadales	Aeromonadaceae	Aeromonas	Aeromonas sobria
353	Proteobacteria	Gammaproteobacteria	Aeromonadales	Aeromonadaceae	Aeromonas	Aeromonas sp. AE235
350	Proteobacteria	Gammaproteobacteria	Enterobacterales	Yersiniaceae	Serratia	Serratia sp. 2015-2462-01
100	Proteobacteria	Gammaproteobacteria	Pseudomonadales	Pseudomonadaceae	Pseudomonas	Pseudomonas sp.1
331	Proteobacteria	Gammaproteobacteria	Pseudomonadales	Pseudomonadaceae	Pseudomonas	Pseudomonas sp.2
333	Proteobacteria	Gammaproteobacteria	Pseudomonadales	Pseudomonadaceae	Pseudomonas	Pseudomonas mandelii
380	Proteobacteria	Gammaproteobacteria	Pseudomonadales	Pseudomonadaceae	Pseudomonas	Pseudomonas peli
322	Proteobacteria	Gammaproteobacteria	Pseudomonadales	Pseudomonadaceae	Pseudomonas	Pseudomonas vancouverensis

Table.2. All the bacteria chosen for this research, with taxonomic information from phylum to species.

2.3. Growth Curve Measuring. The pre-prepared assay plates were then taken to Biotek Synergy 2 plate reader for growth curve measurement. The population of bacteria in each well was proxied by the absorbance (i.e., transparency) at wavelength of 600nm of the solution in the well, as when bacteria multiply the solution gets cloudier due to increasing bacterial cell number. The absorbance of each well of the plates was read and recorded every 1 hour for 72 hours, for all 6 plates of all 21 strains, then plotted into growth curves. All plates had been centrifuged to avoid sedimentation of chemicals. During the measurement of growth curves, strain 85, 398, and 436 did not perform well in terms of population growing and were suspected to be contaminated. This may affect the result of the experiment but there was no solid evidence, and the impact was inferred to be relatively small.

2.4. Growth Curve Data Analysis. Data of all growth curves were stored in the form of absorbance at each hour and corresponding time. The files were imported into RStudio for all analyses in this research (RStudio Team, 2020; R Core Team, 2023). Here I used the area under the curve (AUC) as the index for overall performance of bacteria under the stresses of pesticide chemicals (and DMSO as control condition, since the chemicals were dissolved in DMSO solutions). After AUC values of bacteria cultures being calculated, a Dunnett's test was conducted to test whether each chemical had a significant impact on population growth of the bacteria compared to the control, accounting for multiple tests. The Dunnett's test was performed using R package '*DescTools*' (Signorell, 2023). The chemicals which were suggested significantly affecting growth of at least 1 strain of bacteria were retained for further analyses. Chemicals tested non-significant to all strains would not take part in the remaining analyses. To compare the AUCs between different bacterial species, I used the method in Smith *et al.*'s paper (2023) on bacterial responses to mixtures of chemical stressors with higher order interactions. Relative growth G of bacteria under certain chemical stressor i is defined as the ratio of AUC under the chemical stressor (AUC_i) versus the AUC under the control condition (AUC_{DMSO}):

$$G_i = AUC_i/AUC_{DMSO}$$

For each strain and chemical stressor i, the AUC_i was calculated by averaging AUC values from all three replicates. The AUC ratios were then visualized on a heatmap (Gu *et al.*, 2016), after being $\log_{10}$ transformed. Subsequently, clustering analysis was carried out based on the relative growth data. Note that for all the remaining sections on data analysis and result presentations, Log_{10} transformation is applied to values of G instead of originally calculated AUC ratios.

2.5. Clustering Analysis. Here, to group pesticide chemicals that affect bacteria in similar ways (i.e., to test whether the response of bacteria can be predicted by target groups of pesticide), a Gaussian Mixture Model based clustering analysis on data of G was performed using mclust function from R package *Mclust* (Scrucca *et al.*, 2016). It estimates the parameters of normal mixture models by maximum likelihood estimation via EM (Expectation-maximization) algorithm and fits multivariate data into 14 covariance models with different geometric characteristics (volume, shape, and orientation) (see Scrucca *et al.*, 2016). The number of

components (clusters) was decided by adopting the model with highest (see explanation below) Bayesian Information Criterion (BIC). The formula to calculate BIC is described in the cited *Mclust* version 5 introducing paper above, and stated here:

$$\mathrm{BIC}_{M,G} = 2 \times \ell_{M,G}(\, x \mid \widehat{\Psi}\,) - v \times \log(n)$$

where M and G are the geometric covariance model and number of clusters, ℓ is log-likelihood, $\widehat{\Psi}$ is the maximum likelihood estimation, v is the number of estimated parameters and n is sample size. Note that the calculation of BIC used in this method here actually derives the negative values of BICs that obtained by conventional formula.

To visualize the result of clustering the MclustDR function was applied, which reduces the dimensionality (variable dimensions) of the data and gives plots with coordinates of each 'data point' (in this case the chemicals) at new axes. It is similar to ordination like PCA in terms of dimension reduction and component ranking by the associated eigenvalues but looks to retain most of the cluster structural information (Scrucca, 2010). Result of clustering was plotted and evaluated by Adjusted Rand Index (ARI) to be compared with groups of chemicals by their targets, where the higher the ARI is, the higher the accuracy of the partitioning method is, compared to the original partitioning (Hubert & Arabie, 1985). The ARI is valued between 0 and 1. I have also tried a non-model based hierarchical clustering with the function hclust which produced a qualitatively similar result. For scree plot and clustering result of hclust see appendix.

2.6. Principal Component Analysis (PCA). To further analyse the properties of chemicals by groups, a PCA on AUC ratios of all replicates was performed with chemicals as factor variables. The result of PCA was plotted both on component 1 & 2 as a two-dimensional plane and on component 1 only, due to the eigenvalues of the components. The scale of loadings (lengths of the arguments) was enlarged three times in both axes when plotted to make the graph clearer, otherwise the arrows would be too tiny to observe and look like a cluster of dots.

2.7. Phylogenetic analysis and Mantel test. In this section the hypothesis of positive correlation between phylogenetic resemblance and their phenotypical response to pesticide chemicals was examined. Here I used the 16S rDNA sequences of the 21 strains and aligned the sequences using MAFFT (version 7.520) (Katoh and Standley, 2013; Katoh *et al.*, 2019). Sequentially, the alignment of the examinees was taken to infer a phylogeny tree using RAxML (web-server version 1.0.0) at the default settings with a GTR-gamma substitution model (Kozlov *et al.*, 2019). The phylogeny tree plotting and remaining analyses based on phylogeny tree processing were conducted using the R package '*phytools*' (Revell, 2012).

To primarily observe the relationship between the phenotypic response and phylogeny of each strain, the ancestral states of traits (G in this case) by three different chemicals was reconstructed. The ancestral state estimation was computed by fastAnc function from package *phytools* (Revell, 2012). The three chemicals that

were used to reconstruct ancestral states were randomly selected from the three clusters identified by Mclust respectively. Pagel's lambda is calculated under each of the three chemicals to test the significance of phylogenetic signal (Pagel, 1999). From the positive however ambiguous result of phylogenetic signalling, I decided to carry out a systematic examination on the relationship between phenotypic responses of bacteria to pesticide chemicals and their phylogenetic relatedness using Mantel test with functions from '*vegan*' package (Oksanen *et al.*, 2022). The Mantel test was carried out using pairwise Euclidean distances of *G* between all strains with the phylogenetic distances extracted from the phylogeny tree, in order to evaluate the significance of correlation between the two. Its result, which evidences the predictability of chemical response by evolutionary resemblance, was demonstrated graphically using package '*ggcor*' (Huang *et al.*, 2020).

3. Results

3.1. Growth Curve Analysis. I measured AUCs of 21 strains as overall fitness, in the perspective of population growth, under the stresses of 168 different pesticide chemicals. I then calculated the relative growth *G* of all strains with all chemicals by taking the ratios of AUCs under chemical stresses versus AUCs under control. I assayed the significance of each chemical by running a Dunnett's test and its result is shown as a heatmap below in figure 1 (a), together with the heatmap of *G* values of all 21 strains under chemicals that passed the Dunnett's test (i.e., significantly affecting at least one of the strains) shown in figure 1 (b).

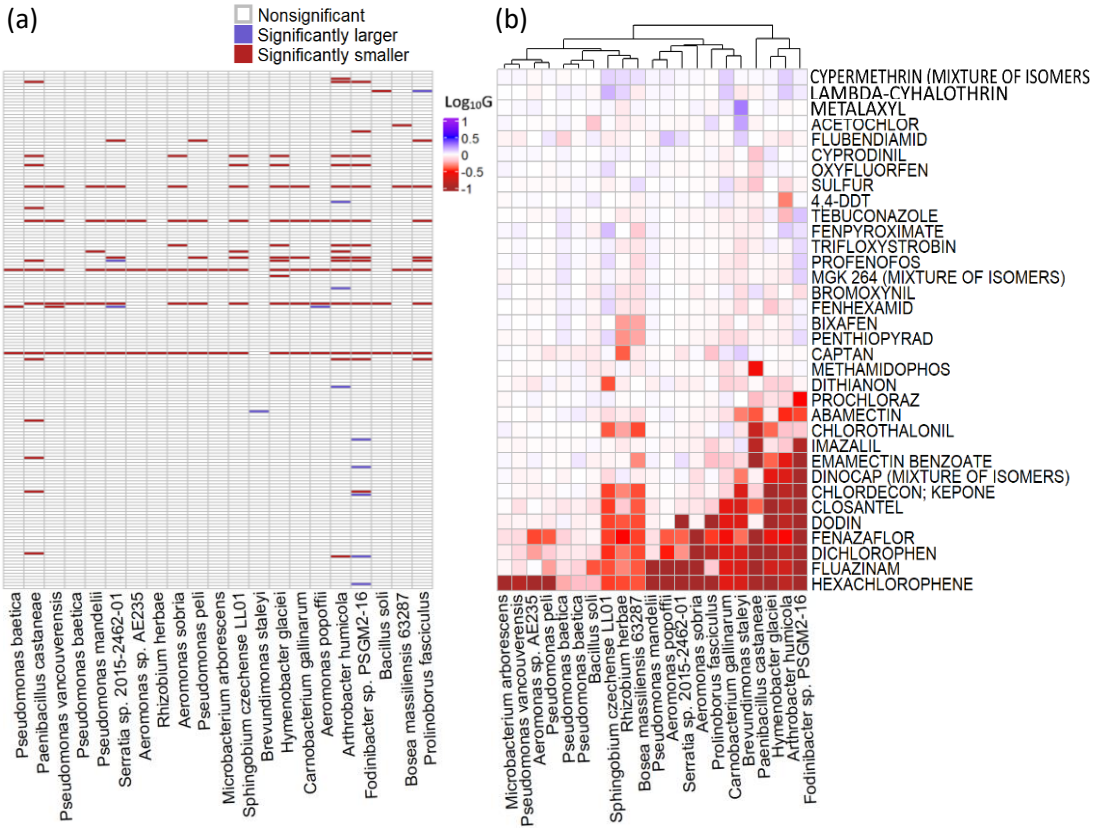

Fig.1. (a) the results of Dunnett's test. Blank cells are *G* values which have no significant change to *G* under control condition. Blue cells are *G* values tested significantly larger, red cells are *G* values tested significantly smaller. Each row in the heatmap represents a unique chemical, each column represents a unique bacterial species. (b) the *G* values of each strain on chemicals that had been tested significant.

From figure 1 (a), vast majority of the chemicals had no significant effect on any of the strains. Thereafter, 134 out of 168 chemicals that are nonsignificant would be shaved from further analyses. In figure 1 (b), relative growth G is shown. The G values were presumed to be mostly distributed around 0, the identical value of G (G_{DMSO}), as I postulated that most of the chemicals have little or no effect on most of the bacteria so that the relative growth of bacteria under majority of the chemicals is close to that of DMSO (control). The substantial cover of white cells in the figure suggests the acceptance of that. Therefore, the blue cells (positive to strain growths) and the red cells (negative to strain growths) can reasonably be used to investigate the phenotypic differences between strains. A hierarchical clustering was automatically performed in the use of Heatmap function and is illustrated in Fig.1 (b). It is also observed that the chemicals are more likely to have negative effect on bacteria than positive effect. Moreover, the negative effects tend to be much stronger than positive effects, if there is any.

3.2. Clustering Analysis. The 34 effective chemicals, in terms of their pesticidal targets, had originally been classified into 7 groups and were analysed by clustering analysis using Mclust. According to model selection by accepting the model with highest BIC value, the geometric covariance model selected is "VVI" with BIC of 1539.739, which clusters the chemicals into 3 groups (for target group of each chemical and detail result of Mclust see appendix). The model name "VVI" stands for variable volumes, variable shapes, and coordinate axial orientations in terms of geometric distributions of different clusters. The visualization of this result is illustrated in figure 2 where Dir1 and Dir2 among the reduced dimensions by MclustDR are the two most informative axes in terms of cluster structures. The chemicals are represented by the data points scattered on the coordinate frame of Dir1 and Dir2 and drawn in different colours by their clusters. From figure 2 there is no clear correlation between clusters by Mclust and by target groups observed. The plots on the top and right-hand side are the density plots (scaled to maximum of 1) of the clusters along Dir1 and Dir2 respectively. Compared to cluster 1 and 2, cluster 3 is more intensively distributed, which means the data points of cluster 3 are relatively closer to each other. The number of chemicals partitioned into cluster 3 is also smaller. Furthermore, along the axis Dir1, the most informative dimension, cluster 3 is substantially further away from central point 0 than cluster 1 and 2, although cluster 1 and 2 are distributed oppositely along Dir 2. The comparison table of the two partitions is shown in table 3, an obvious 'group-crossing' is observed where members from one group by grouping method 1 cross various groups by grouping method 2. The ARI (Adjusted Rand Index, as stated in methodology) is 0.036428 which is low enough to suggest that the two partitions by target organisms and by Mclust are independent.

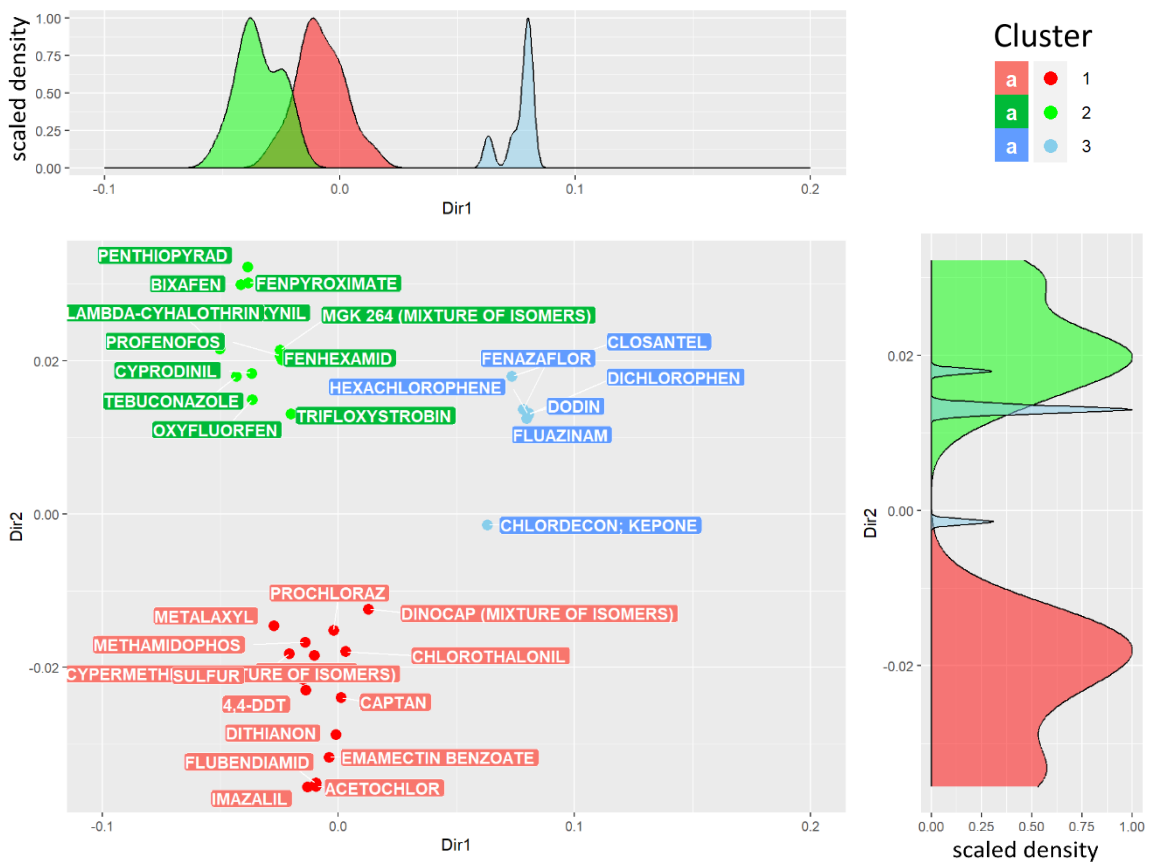

Fig.2. The dimension reduction visualization of Mclust result using MclustDR function. The new axes Dir1 and Dir2 are the most cluster-informative variables. Each data point on the graph represents a chemical. The chemicals are classified into three clusters, illustrated using three different colours as shown in the plot at top right corner. The plots on the top and right-hand side are the density plots (scaled to maximum of 1) of the chemicals on the two axes.

Cluster	Fungicide	Herbicide	Insecticide	Insecticide, Acaricide	Multi-target	Parasiticide	Synergist
Cluster 1	7	1	6	0	1	0	0
Cluster 2	6	2	2	1	0	0	1
Cluster 3	2	0	1	1	2	1	0

Table.3. The total of 34 chemicals partitioned into 3 clusters by Mclust and 7 groups by disinfecting target. There is a clear 'group-crossing' in the 34 chemicals.

3.3. Principal Component Analysis. Thus, according to the result of clustering, the target groups of pesticide chemicals and their effects on bacteria are independent between each other and have no correlation, which evidences the rejection of the first hypothesis. To verify the result and to get a deeper look at the impact tendencies of different clusters of chemicals on bacteria, I additionally carried out a PCA on all the replicates of bacteria strains. The scree plot for eigenvalues of each component is in appendix, in which only the component 1 has eigenvalue larger than one. Therefore, although I still have plotted a common PCA graph, I also attached the plot with only component 1 to figure 3. The loadings and corresponding labels are in different colours by the clusters they belong to, from the results of clustering. In Fig.3 (a) it is observed that the loadings

of PCA are clearly distinguished by relatively different arguments and directions, which volumes up the partitioning of Mclust. However, on Fig.3 (b), the one-dimensional graph, it shows that almost all the chemicals have loadings towards positive direction and the clusters are only distinguished by the length of loadings. Even the very few chemicals that load on negative part of PC1 have very small arguments. In concert with the result of clustering, chemicals of cluster 3 tend to have much stronger positive loadings along PC1.

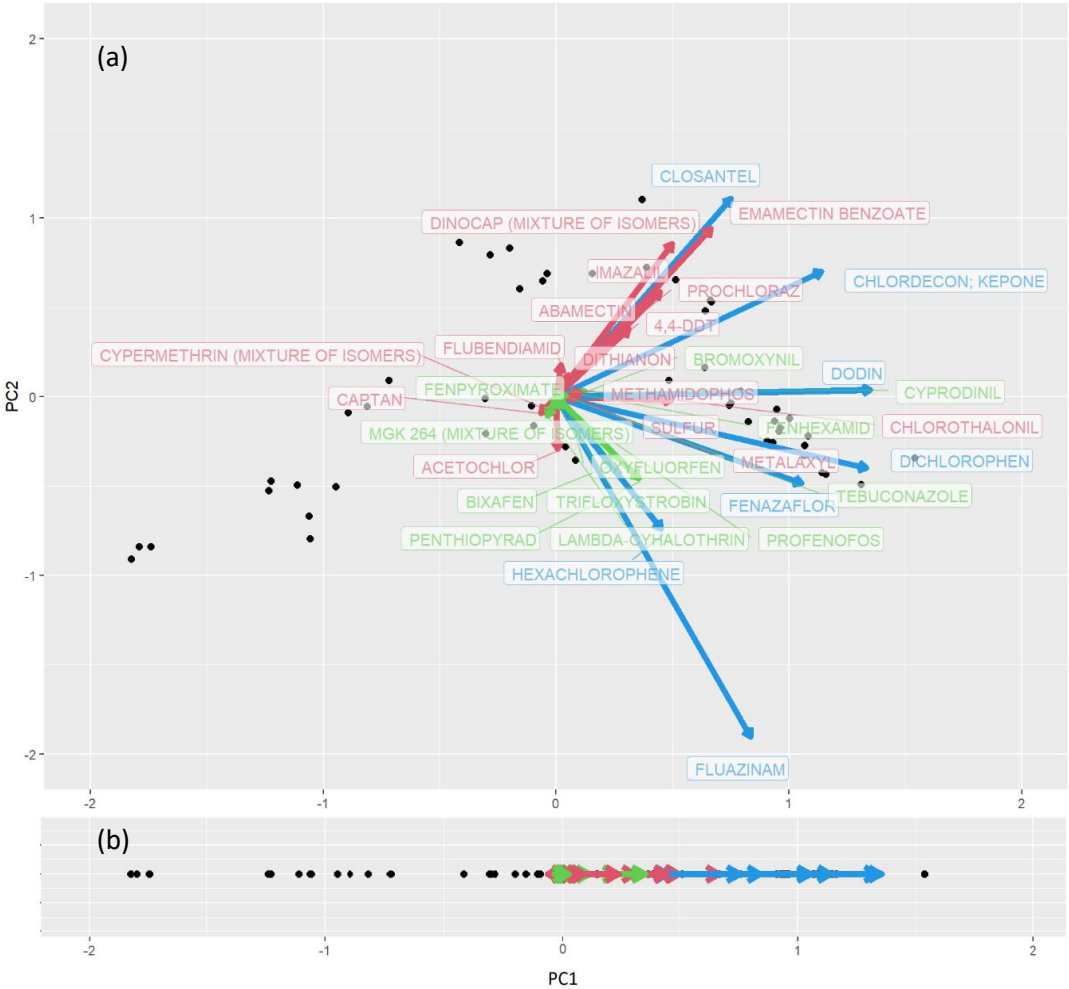

Fig.3. The plots of PCA result. The loadings are differentiated by colours. The colour of each group of chemicals are in constant with that in Fig.2, in which red is cluster 1, green is cluster 2, and blue is cluster 3. The scale of the loadings (lengths of the arrows) is three times enlarged to better demonstrate the difference between loadings.

3.4. Phylogenetic Analysis. In this section, I tested the relationship between phenotypic responses of bacteria and their phylogenetic relatedness. Overall, it is suggested by the result of this section that the response of bacteria to pesticide chemicals is predictable by examining their evolutionary resemblance. The basic phylogeny tree built from the alignment of 16S rDNA sequences of the strains is shown in the figure 5 (d), together with the result of Mantel test. The ancestral states of *G* under three chemical stressors randomly selected from the three distinct clusters by Mclust were reconstructed. The three chemicals chosen are Abamectin, Fenpyroximate, and Dichlorophen, from cluster 1, 2, and 3 respectively. The phylogeny trees with ancestral states of traits projected, together with plots of phylomorphospace (two-dimensional plot of two continuous bacterial traits. In this case, *G* under stressors) of *G* under the three chemicals were combined into

a graph that is called a 'phylogenetic scatterplot matrix' by the author of package '*PhyTools*'. The phylogenetic scatterplot matrix is demonstrated in figure 4. It is observed that in most circumstances the species that are closely related in phylogenetic perspective have similar responses to each of the chosen chemical stress. In the phylogeny trees, closer species tend to have similar colours (trait values) and in the plots of phylomorphospace closer species are more likely to gather with each other. Be aware that the colours shown in the legends in each of the ancestral state plots have different *G* values indicated. I then calculated the Pagel's lambdas of the three traits (chemicals) as a measure of phylogenetic signal, and their statistical significances. The Pagel's lambdas of phylogenetic signal are 0.528914, 0.999927, 0.244962 and the p-values are 0.0327745, 0.0350435, 0.429555, respectively to Abamectin, Fenpyroximate, and Dichlorophen. Pagel's lambda is an indication of phylogenetic signal as mentioned in Methodology, which value of 0 infers no phylogenetic signal (i.e., traits evolved independently from phylogeny) of the traits and value of 1 corresponds to a Brownian Motion model of evolution. Note that in this calculation of Pagel's lambdas here (and here only), ratios of AUC before $\log_{10}$ transformed were used. The phylogenetic signal is significant for Abamectin and Fenpyroximate but not for Dichlorophen. From the result I suspected that there is a strong phylogenetic signal of the species, but further analysis was required.

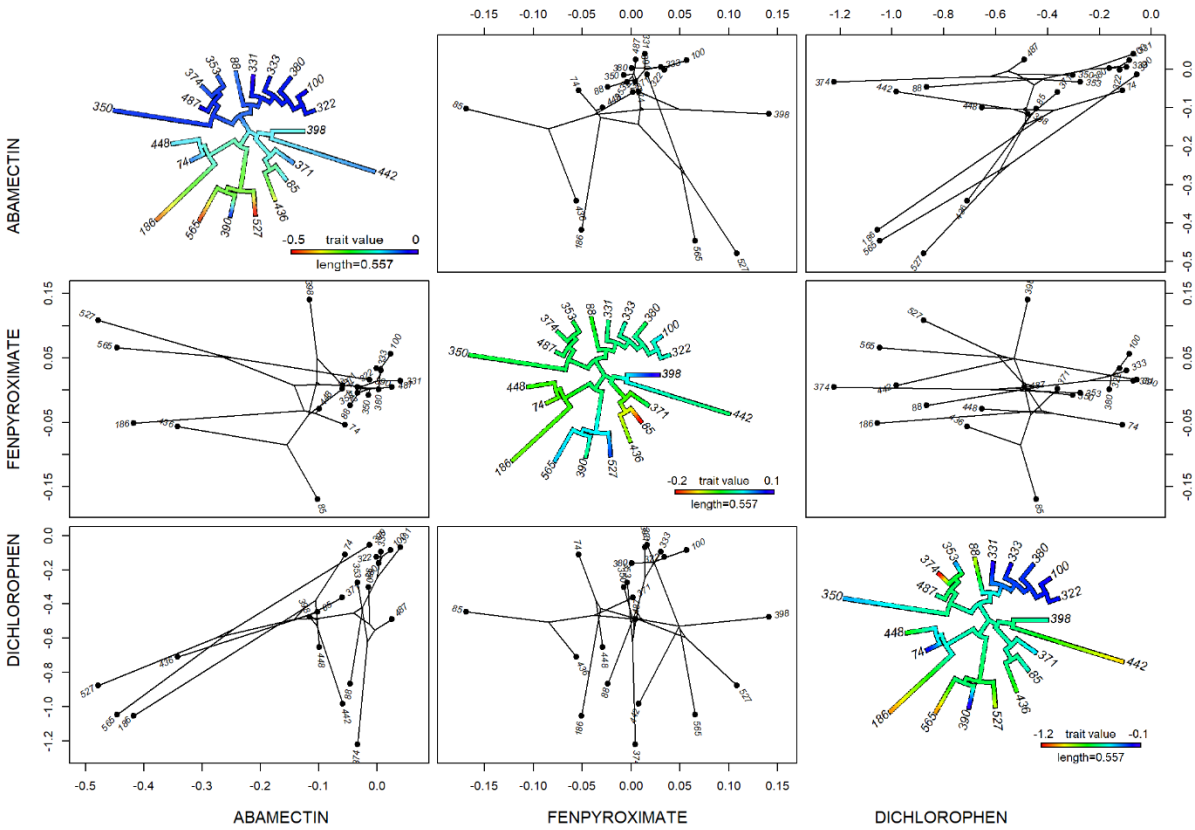

Fig.4. The phylogenetic scatterplot matrix of G under Abamectin, Fenpyroximate, and Dichlorophen. The axes of the phylomorphospace plots (graphs off diagonal) are the values of traits (G under the chemicals) of bacteria under the stresses of labelled chemicals. The plots at diagonal are the reconstructions of ancestral states of G under the chemicals.

3.5. Mantel Test. Considering the result of strong phylogenetic signal with somewhat ambiguity, I ultimately applied a Mantel test using Euclidean distances extracted from the phylogeny tree and the distance matrix of *G*. The correlation between phylogenetic distance and phenotypic response (*G* under chemical stresses) was tested remarkably significant with Mantel statistic r of 0.3373 and p-value of 5e-04. To solid the result, I ran another Mantel test based on data of all 168 chemicals and there was no noteworthy difference in the outcomings (r = 0.3584, p-value = 5e-04). The result of Mantel test was plotted on figure 5, along with the phylogeny tree.

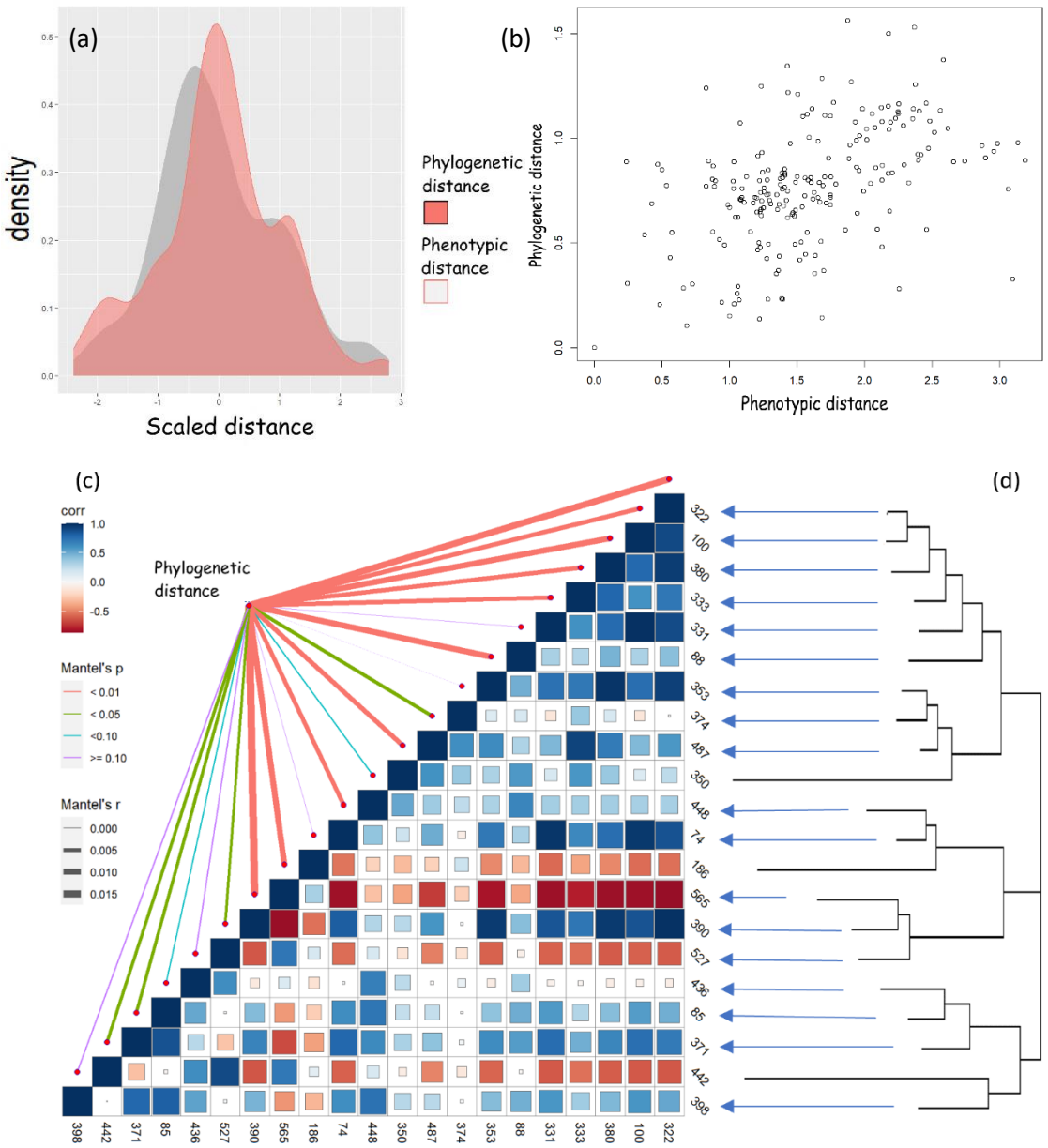

Fig.5. (a) The density plot of scaled phenotypic and phylogenetic distances. (b) Phylogenetic distance against pair wise phenotypic (*G*) distance. (c) The detailed result of Mantel test. Correlation coefficient of phylogenetic distance to phenotypic distance of each single strain is represented by linewidth, their corresponding p-values are categorized in different colours. Correlation of *G* (phenotypic) between strains is shown as the colour and size of each square in the bottom part. (d) The phylogeny tree built by RAxML using alignment of 16S rDNA.

The densities of phenotypic distance and phylogenetic distance extracted from the phylogeny tree are plotted in figure 5 (a), with a highly overlapped area. Figure 5 (b) shows the raw data of phylogenetic and phenotypic distances for the Mantel test. A clear positive correlation can be observed. In figure 5 (c), lines of varying widths stand for the correlation coefficients of phylogenetic distance and phenotypic distance, specific to each strain. The colours of lines represent the p-values. The colours and sizes of the square bricks are the correlation of *G* between the 21 strains of bacteria. In terms of statistical significance, 10 out of 21 strains have phenotypic responses to pesticide chemicals firmly correlated with phylogenetic distance (p-value < 0.01), 4 strains have correlation conventionally considered significant ($0.01 < p < 0.05$). If rejection area is 0.05, there are 7 strains that do not have phenotypic response significantly correlated with phylogenetic relatedness, but if the rejection area is lowered to 0.10, that number of strains will be lowered to 5. Among the 7 strains tested nonsignificant, strain 398, 85, and 436 were suspected contaminated during the measurement of AUC, leaving only 4 strains of bacteria suspected exceptional.

4. Discussion

4.1. Environmental Impact Perspective. From the results of analysis on bacteria relative growths, we had a general idea on how the chosen bacterial species react to the 168 tested chemicals. In the perspective of bacteria all the chosen strains showed significant variation from control in response to stresses of least one chemical, where most of the responses showed negative effects, while some previous studies had shown contradictory results possibly due to less and different bacteria and chemicals or different concentrations of chemicals used in experiments (Staley *et al.*, 2015; Astaykina *et al.*, 2020). Not only negative but also positive effects on bacteria were detected, in line with results from previous research on bioremediation where some bacteria were proved able to break down and metabolize on some pesticide chemicals (e.g., Geetha and Fulekar, 2008; Odukkathil and Vasudevan, 2013). However, the negative effects on tested strains are much stronger and more frequent than positive effects, which could mean that most bacterial species may experience much stronger impact from most pesticide chemicals compared to beneficial non-target effects. This may be explained by destructional properties of chemicals at cellular or biochemical level or nutrient inhibition as concluded by Shahid and Khan (2022). On the other hand, in the perspective of pesticide chemicals, majority of the chemicals used in this experiment which are pesticide ingredients had no significant effect on the tested bacteria strains, which means the impact of pesticide leaching may not be as harmful as we previously presumed. This result may be affected by using different dosages of toxic chemicals as I have only tested 1 dosage of each chemical here, in order to do a large screening of different chemicals and bacterial strains within the allowed research duration. Combining the two points of view it is suggested that although bacteria tend to be negatively affected by pesticide chemicals, the responses are highly stressor specific and concentration dependent. Furthermore, the results of phylogenetic analysis and Mantel test suggest that the impact of chemicals on certain bacterial species is predictable by the phylogenetic relatedness of the bacteria. It is useful when predict the ecological impact of the use of certain pesticides, however, further study is needed as I have

only tested monoculture bacteria in a nutrient rich medium. In real world ecosystems, bacteria usually appear as communities that interact internally and are nutrient limited, the pesticide pollutants are also commonly mixtures of chemicals (Stubbendieck *et al.*, 2016; Bell, 2019; Smith *et al.*, 2023).

4.2. Biosensor Perspective. Both the results of clustering analysis and PCA showed no evidence for similar effects of chemicals with same target organisms. The Mclust result demonstrated that the chemicals were partitioned into three groups by their effects on bacteria while originally partitioned into 7 groups by their disinfectant targets, where the two partitions match terribly. There has been very few research on how chemicals within a same group may impact distinctly on microbes. Therefore, further research is needed to evidence the finding. The PCA gave a similar result with the same conclusion. The result of PCA here shows that the partitions made by Mclust are obviously distinct. In concert with the findings of relative growth analysis and Mclust, the result of PCA shows that most of the chemicals do tend to affect bacteria in one direction (Positive along PC1. Presumed to be negative effects here, as all the cluster 3 chemicals which caused strong negative effects according to figure.1 lie positively along PC1 and to a much greater extent than the other two clusters) and the effects in this direction are much stronger than the opposite. The reason to run Mclust and PCA simultaneously is that although Mclust as a partitioning analysis gives a classification of the tested chemicals and reveals the relationship between itself and groups by targets, it does not show particularly the impact of a certain chemical on bacteria. The PCA does not provide a clear classification of the chemicals, but it does provide a deeper inspection on how each of the chemicals affects bacteria specifically and the result of PCA clearly consolidated the outcomings of clustering analysis. Thereafter, the first hypothesis is obviously rejected.

Contrarily, the second hypothesis, that phylogenetically similar bacteria have similar responses to pesticide chemicals, is accepted with a vigorously significant result. The phylogenetic signal was tested significant for 2 out of 3 chemicals chosen from different clusters by Mclust. The Mantel test showed a powerfully significant correlation between the bacteria responses to chemicals and the phylogenetic relatedness of the bacteria. This contrasts with the result of previous experiment in mixture-chemical scenarios by Smith *et al.* (2023).

In terms of biosensors, results of both hypotheses give useful clues for future studies: **1.** Although the chemicals do not affect bacteria collectively by their groups of target organisms, they do cluster into three different partitions. In further research on different chemicals, methods such as discriminant analysis can be used to identify the clusters those chemicals belonging to, and it would be possible to predict their impacts on the same bacterial species. **2.** From the result of Mantel test, the significant correlation between bacterial responses and phylogenetic distances provides a convenient way to infer the responses of certain bacteria to pesticide chemicals based on known responses of phylogenetically resembling bacteria strains. Phylogenetically related bacterial species and their similarity in chemical responses have not been thoroughly studied, but their ecological performances in some other respects had been found related (e.g., Morrissey and Franklin, 2015). However, I suggest that further studies and evidence are needed to verify the feasibility of this method. More strains of bacteria should have been tested but abandoned due to tightened experiment schedule. Moreover, although bacteria have been proved acting similarly in many ecological ways, the use of bacterial phylogeny

has been debated (Morrissey *et al.*, 2016). The use of bacterial 16S rDNA sequences had also been suspected unreliable (Jaspers and Overmann, 2004). Despite all, the potential benefits of further understanding this relationship between bacterial phylogeny and their responses to pesticide chemicals and exploring the applicability of this predicting method is still worth trying.

5. Conclusion

This research reveals the phenotypic responses of 21 strains of bacteria to 168 different pesticide chemicals. The bacteria react actively to chemicals, in both positive and negative way, while most of the chemicals had no significant impact on bacteria. The negative effects on bacteria are much stronger and more frequent than positive effects. By clustering analysis and PCA, I found that chemicals of same group of target organisms do not affect bacteria in similar way and have rejected first hypothesis. However, similar bacteria in terms of phylogenetic distance showed significant similarity in responses to toxic pesticide chemicals. Therefore, second hypothesis was accepted. Such results can be indicative for further studies on biosensors of pesticide pollutants.

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Word count: 5261

All raw data and R code used in this research can be found here:

https://github.com/skymanbp/EEC_01740950_Project_code.git

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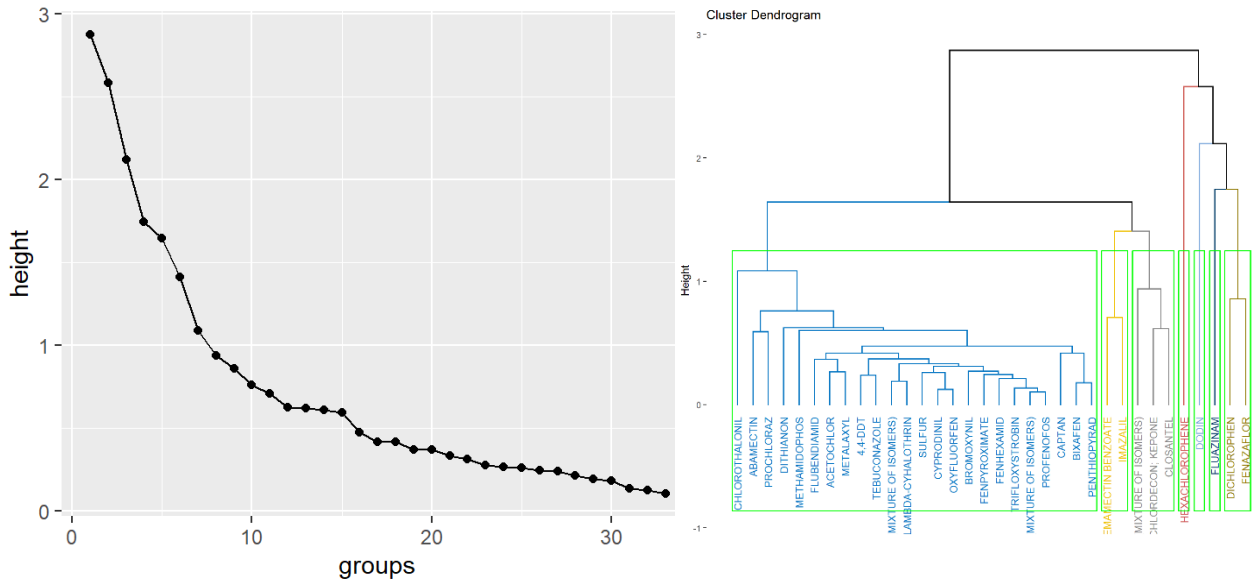
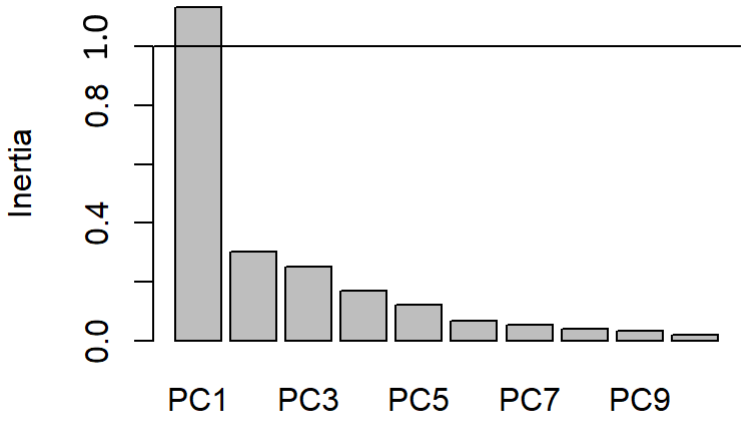
Appendix.

1. The scree plot (left) and result (right) for hierarchical clustering using hclust function.



2. Table showing the group of each chemical belonging to, by both target organism and Mclust, and corresponding coordinates on Dir1 and Dir 2 by MclustDR function.

Chemical	Cluster	Target	Dir1	Dir2
4,4-DDT	1	Insecticide	-0.01397	-0.02295
ABAMECTIN	1	Insecticide	-0.01017	-0.01846
ACETOCHLOR	1	Herbicide	-0.0095	-0.03549
BIXAFEN	2	Fungicide	-0.04128	0.029909
BROMOXYNIL	2	Herbicide	-0.03366	0.02637
CAPTAN	1	Fungicide	0.001066	-0.02396
CHLORDECON; KEPONE	3	Insecticide	0.063075	-0.00145
CHLOROTHALONIL	1	Fungicide	0.003017	-0.01794
CLOSANTEL	3	Parasiticide	0.073509	0.017961
CYPERMETHRIN (MIXTURE OF ISOMERS)	1	Insecticide	-0.02084	-0.01818
CYPRODINIL	2	Fungicide	-0.03668	0.018368
DICHLOROPHEN	3	Multi-target	0.080486	0.012935
DINOCAP (MIXTURE OF ISOMERS)	1	Fungicide	0.012746	-0.0124
DITHIANON	1	Fungicide	-0.00106	-0.02874
DODIN	3	Fungicide	0.080589	0.013112
EMAMECTIN BENZOATE	1	Insecticide	-0.00403	-0.03174
FENAZAFLOR	3	Insecticide, Acaricide	0.078314	0.013698
FENHEXAMID	2	Fungicide	-0.02409	0.020151
FENPYROXIMATE	2	Insecticide, Acaricide	-0.03812	0.03012
FLUAZINAM	3	Fungicide	0.07981	0.012447
FLUBENDIAMID	1	Insecticide	-0.00953	-0.0351
HEXACHLOROPHENE	3	Multi-target	0.080223	0.013044
IMAZALIL	1	Fungicide	-0.01318	-0.03557
LAMBDA-CYHALOTHRIN	2	Insecticide	-0.05028	0.021538
METALAXYL	1	Fungicide	-0.02746	-0.01458
METHAMIDOPHOS	1	Insecticide	-0.01419	-0.01672
MGK 264 (MIXTURE OF ISOMERS)	2	Synergist	-0.02474	0.021422
OXYFLUORFEN	2	Herbicide	-0.03642	0.014918
PENTHIOPYRAD	2	Fungicide	-0.03857	0.032264
PROCHLORAZ	1	Fungicide	-0.0021	-0.01517
PROFENOFOS	2	Insecticide	-0.02467	0.020647
SULFUR	1	Multi-target	-0.01499	-0.02152
TEBUCONAZOLE	2	Fungicide	-0.04319	0.017983
TRIFLOXYSTROBIN	2	Fungicide	-0.02014	0.013064

3. The scree plot for components of PCA